

8 RULES OF IMPLICATION

1. Modus Ponens (MP)

$$\begin{array}{l} p \supset q \\ p \\ \therefore q \end{array}$$

2. Modus Tollens (MT)

$$\begin{array}{l} p \supset q \\ \sim q \\ \therefore \sim p \end{array}$$

3. Hypothetical Syllogism (HS)

$$\begin{array}{l} p \supset q \\ q \supset r \\ \therefore p \supset r \end{array}$$

4. Disjunctive Syllogism (DS)

$$\begin{array}{l} p \vee q \\ \sim p \\ \therefore q \end{array}$$

5. Constructive Dilemma (CD)

$$\begin{array}{l} (p \supset q) \cdot (r \supset s) \\ p \vee r \\ \therefore q \vee s \end{array}$$

6. Simplification (Simp)

$$\begin{array}{l} p \cdot q \\ \therefore p \end{array}$$

7. Conjunction (Conj)

$$\begin{array}{l} p \\ q \\ \therefore p \cdot q \end{array}$$

8. Addition (Add)

$$\begin{array}{l} p \\ \therefore p \vee q \end{array}$$

10 RULES OF REPLACEMENT

9. De Morgan's Theorems (DM)

$$\begin{array}{ll} \sim (p \cdot q) & :: \sim p \vee \sim q \\ \sim (p \vee q) & :: \sim p \cdot \sim q \end{array}$$

10. Association (Assoc)

$$\begin{array}{ll} p \vee (q \vee r) & :: (p \vee q) \vee r \\ p \cdot (q \cdot r) & :: (p \cdot q) \cdot r \end{array}$$

11. Double Negation (DN)

$$p :: \sim \sim p$$

12. Material Implication (Impl)

$$p \supset q :: \sim p \vee q$$

13. Exportation (Exp)

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

14. Commutation (Com)

$$\begin{array}{ll} p \vee q & :: q \vee p \\ p \cdot q & :: q \cdot p \end{array}$$

15. Distribution (Dist)

$$\begin{array}{ll} p \cdot (q \vee r) & :: (p \cdot q) \vee (p \cdot r) \\ p \vee (q \cdot r) & :: (p \vee q) \cdot (p \vee r) \end{array}$$

16. Transposition (Trans)

$$p \supset q :: \sim q \supset \sim p$$

17. Material Equivalent (Equiv)

$$\begin{array}{ll} p \equiv q & :: (p \supset q) \cdot (q \supset p) \\ p \equiv q & :: (p \cdot q) \vee (\sim p \cdot \sim q) \end{array}$$

18. Tautology (Taut)

$$\begin{array}{ll} p & :: p \vee p \\ p & :: p \cdot p \end{array}$$